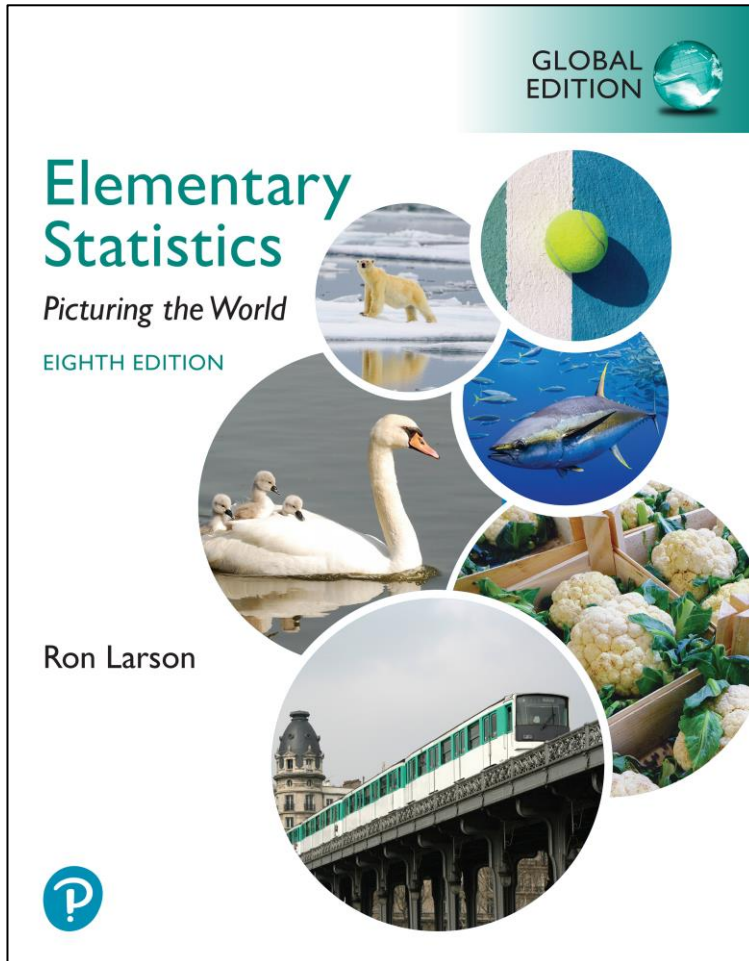


Elementary Statistics

Eighth Edition



Chapter 4

Discrete Probability Distributions

Chapter Outline

4.1 Probability Distributions

4.2 Binomial Distributions

4.3 More Discrete Probability Distributions

Section 4.2 Binomial Distributions

Section 4.2 Objectives

1. How to determine whether a probability experiment is a binomial experiment
2. How to find binomial probabilities using the binomial probability formula
3. How to find binomial probabilities using technology, formulas, and a binomial probability table
4. How to construct and graph a binomial distribution
5. How to find the mean, variance, and standard deviation of a binomial probability distribution

Binomial Experiments

1. The experiment is repeated for a fixed number of trials, where each trial is independent of other trials.
2. There are only two possible outcomes of interest for each trial. The outcomes can be classified as a success (S) or as a failure (F).
3. The probability of a success, $P(S)$, is the same for each trial.
4. The random variable x counts the number of successful trials.

Notation for Binomial Experiments

Symbol	Description
n	The number of times a trial is repeated
p	The probability of success in a single trial
q	The probability of failure in a single trial ($q = 1 - p$)
x	The random variable represents a count of the number of successes in n trials: $x = 0, 1, 2, 3, \dots, n$.

Example: Identifying and Understanding Binomial Experiments (1 of 2)

Decide whether each experiment is a binomial experiment. If it is, specify the values of n , p , and q , and list the possible values of the random variable x . If it is not, explain why.

1. A certain surgical procedure has an 85% chance of success. A doctor performs the procedure on eight patients. The random variable represents the number of successful surgeries.

Solution: Identifying and Understanding Binomial Experiments (1 of 3)

Binomial Experiment

1. Each surgery represents a trial. There are eight surgeries, and each one is independent of the others.
2. There are only two possible outcomes of interest for each surgery: a success (S) or a failure (F).
3. The probability of a success, $P(S)$, is 0.85 for each surgery.
4. The random variable x counts the number of successful surgeries.

Solution: Identifying and Understanding Binomial Experiments (2 of 3)

Binomial Experiment

- $n = 8$ (number of trials)
- $p = 0.85$ (probability of success)
- $q = 1 - p = 1 - 0.85 = 0.15$ (probability of failure)
- $x = 0, 1, 2, 3, 4, 5, 6, 7, 8$ (number of successful surgeries)

Example: Identifying and Understanding Binomial Experiments (2 of 2)

Decide whether each experiment is a binomial experiment. If it is, specify the values of n , p , and q , and list the possible values of the random variable x . If it is not, explain why.

2. A jar contains five red marbles, nine blue marbles, and six green marbles. You randomly select three marbles from the jar, without replacement. The random variable represents the number of red marbles.

Solution: Identifying and Understanding Binomial Experiments (3 of 3)

Not a Binomial Experiment

- The probability of selecting a red marble on the first trial is $\frac{5}{20}$.
- Because the marble is not replaced, the probability of success (red) for subsequent trials is no longer $\frac{5}{20}$.
- The trials are not independent and the probability of a success is not the same for each trial.

Binomial Probability Formula

Binomial Probability Formula

- The probability of exactly x successes in n trials is

$$P(x) = {}_n C_x p^x q^{n-x} = \frac{n!}{(n-x)! x!} p^x q^{n-x}$$

- n = number of trials
- p = probability of success
- $q = 1 - p$ probability of failure
- x = number of successes in n trials
- Note: number of failures is $n - x$

Example: Finding a Binomial Probability

Rotator cuff surgery has a 90% chance of success. The surgery is performed on three patients. Find the probability of the surgery being successful on exactly two patients. (Source: The Orthopedic Center of St. Louis)



Solution: Finding a Binomial Probability (1 of 2)

Method 1: Draw a tree diagram and use the Multiplication Rule.

1st Surgery	2nd Surgery	3rd Surgery	Outcome	Number of Successes	Probability
S	S	S	SSS	3	$\frac{9}{10} \cdot \frac{9}{10} \cdot \frac{9}{10} = \frac{729}{1000}$
		F	SSF	2	$\frac{9}{10} \cdot \frac{9}{10} \cdot \frac{1}{10} = \frac{81}{1000}$
	F	S	SFS	2	$\frac{9}{10} \cdot \frac{1}{10} \cdot \frac{9}{10} = \frac{81}{1000}$
		F	SFF	1	$\frac{9}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} = \frac{9}{1000}$
	S	S	FSS	2	$\frac{1}{10} \cdot \frac{9}{10} \cdot \frac{9}{10} = \frac{81}{1000}$
		F	FSF	1	$\frac{1}{10} \cdot \frac{9}{10} \cdot \frac{1}{10} = \frac{9}{1000}$
F	F	S	FFS	1	$\frac{1}{10} \cdot \frac{1}{10} \cdot \frac{9}{10} = \frac{9}{1000}$
		F	FFF	0	$\frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} = \frac{1}{1000}$

$$3 \left(\frac{81}{1000} \right) = 0.243$$

Solution: Finding a Binomial Probability (2 of 2)

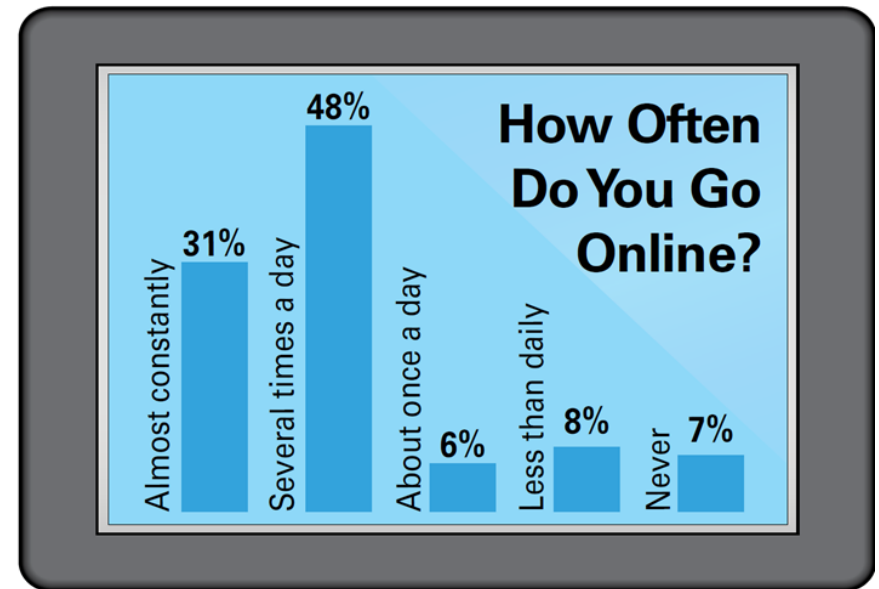
Method 2: Use the binomial probability formula.

$$n = 3, p = \frac{9}{10}, q = \frac{1}{10}, \text{ and } x = 2.$$

$$\begin{aligned} P(2) &= \frac{3!}{(3-2)!2!} \left(\frac{9}{10}\right)^2 \left(\frac{1}{10}\right)^1 \\ &= 3 \left(\frac{81}{100}\right) \left(\frac{1}{10}\right) \\ &= 3 \left(\frac{81}{1000}\right) \\ &= 0.243 \end{aligned}$$

Example: Constructing a Binomial Distribution

In a survey, U.S. adults were asked how often they go online. The results are shown in the figure. Six adults who participated in the survey are randomly selected and asked whether they go **online several times a day**. Construct a binomial probability distribution for the number who respond that they go online several times a day. (Source: Pew Research)



Solution: Constructing a Binomial Distribution (1 of 2)

$$p = 0.48 \text{ and } q = 0.52$$

$n = 6$, possible values for x are 0, 1, 2, 3, 4, 5 and 6

$$P(0) = {}_6C_0 (0.48)^0 (0.52)^6 = 1(0.48)^0 (0.52)^6 \approx 0.020$$

$$P(1) = {}_6C_1 (0.48)^1 (0.52)^5 = 6(0.48)^1 (0.52)^5 \approx 0.109$$

$$P(2) = {}_6C_2 (0.48)^2 (0.52)^4 = 15(0.48)^2 (0.52)^4 \approx 0.253$$

$$P(3) = {}_6C_3 (0.48)^3 (0.52)^3 = 20(0.48)^3 (0.52)^3 \approx 0.311$$

$$P(4) = {}_6C_4 (0.48)^4 (0.52)^2 = 15(0.48)^4 (0.52)^2 \approx 0.215$$

$$P(5) = {}_6C_5 (0.48)^5 (0.52)^1 = 6(0.48)^5 (0.52)^1 \approx 0.079$$

$$P(6) = {}_6C_6 (0.48)^6 (0.52)^0 = 1(0.48)^6 (0.52)^0 \approx 0.012$$

Solution: Constructing a Binomial Distribution (2 of 2)

Notice in the table that all the probabilities are between 0 and 1 and that the sum of the probabilities is about 1.

x	$P(x)$
0	0.020
1	0.109
2	0.253
3	0.311
4	0.215
5	0.079
6	0.012
	$\sum P(x) = 0.999$

Example: Finding a Binomial Probabilities Using Technology

A survey found that 75% of U.S. adults are experiencing digital device fatigue. (This is the state of mental exhaustion caused by concurrent and excessive use of digital devices such as smartphones and tablet computers.) You randomly select 100 adults. What is the probability that exactly 66 are experiencing digital device fatigue? Use technology to find the probability. (Source: The Harris Poll)

Solution: Finding a Binomial Probabilities Using Technology (1 of 2)

Solution

Minitab, Excel, StatCrunch, and the TI-84 Plus each have features that allow you to find binomial probabilities. Try using these technologies. You should obtain results similar to these displays.

MINITAB

Probability Density Function

Binomial with $n = 100$ and $p = 0.75$

x	P(X = x)
66	0.0111678

STATCRUNCH

Binomial Calculator

n:100 p:0.75
 $P(X = 66) = 0.01116782$

TI-84 PLUS

binompdf(100, .75, 66)
.0111678235

EXCEL

	A
1	0.011167823

←=BINOM.DIST(66,100,0.75,FALSE)

Solution: Finding a Binomial Probabilities Using Technology (2 of 2)

Solution

From these displays, you can see that the probability that exactly 66 adults are experiencing digital device fatigue is about 0.011. Because 0.011 is less than 0.05, this can be considered an unusual event.

Example: Finding Binomial Probabilities Using Formulas

A survey found that 22% of U.S. adults say that the economy, unemployment, and jobs are the most important problems facing the U.S. today. You randomly select four adults and ask them whether the economy, unemployment, and jobs are the most important problems facing the U.S. today. Find the probability that (1) exactly two of them respond yes, (2) at least two of them respond yes, and (3) fewer than two of them respond yes. (Source: Ipsos)

Solution: Finding Binomial Probabilities Using Formulas (1 of 4)

Solution

1. Using $n = 4$, $p = 0.22$, $q = 0.78$, and $x = 2$, the probability that exactly two adults will respond yes is

$$P(2) = {}_4C_2 (0.22)^2 (0.78)^2 = 6(0.22)^2 (0.78)^2 \approx 0.177.$$

Solution: Finding Binomial Probabilities Using Formulas (2 of 4)

Solution

2. To find the probability that at least two adults will respond yes, find the sum of $P(2)$, $P(3)$, and $P(4)$.

Begin by using the binomial probability formula to write an expression for each probability.

$$P(2) = {}_4C_2 (0.22)^2 (0.78)^2 = 6(0.22)^2 (0.78)^2$$

$$P(3) = {}_4C_3 (0.22)^3 (0.78)^1 = 4(0.22)^3 (0.78)^1$$

$$P(4) = {}_4C_4 (0.22)^4 (0.78)^0 = 1(0.22)^4 (0.78)^0$$

Solution: Finding Binomial Probabilities Using Formulas (3 of 4)

Solution

2. So, the probability that at least two will respond yes is

$$\begin{aligned} P(x \geq 2) &= P(2) + P(3) + P(4) = 6(0.22)^2(0.78)^2 \\ &\quad + 4(0.22)^3(0.78)^1 + 1(0.22)^4(0.78)^0 \\ &\approx 0.212. \end{aligned}$$

Solution: Finding Binomial Probabilities Using Formulas (4 of 4)

Solution

3. To find the probability that fewer than two adults will respond yes, find the sum of $P(0)$ and $P(1)$.

$$P(0) = {}_4C_0 (0.22)^0 (0.78)^4 = 1(0.22)^0 (0.78)^4$$

$$P(1) = {}_4C_1 (0.22)^1 (0.78)^3 = 4(0.22)^1 (0.78)^3$$

So, the probability that fewer than two will respond yes is

$$\begin{aligned} P(x < 2) &= P(0) + P(1) \\ &= 1(0.22)^0 (0.78)^4 + 4(0.22)^1 (0.78)^3 \approx 0.788. \end{aligned}$$

Example: Finding a Binomial Probability Using a Table

About 5% of workers (ages 16 years and older) in the United States commute to their jobs by using public transportation (excluding taxicabs). You randomly select eight workers. What is the probability that exactly three of them use public transportation to get to work? Use a table to find the probability. (Source: American Community Survey)

Solution:

- Binomial with $n = 8$, $p = 0.05$, $x = 3$

Solution: Finding Binomial Probabilities Using a Table (1 of 2)

- A portion of Table 2 is shown (Appendix B Table 2 – Binomial Distribution)

		<i>p</i>												
<i>n</i>	<i>x</i>	.01	.05	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60
2	0	.980	.902	.810	.723	.640	.563	.490	.423	.360	.303	.250	.203	.160
	1	.020	.095	.180	.255	.320	.375	.420	.455	.480	.495	.500	.495	.480
	2	.000	.002	.010	.023	.040	.063	.090	.123	.160	.203	.250	.303	.360
3	0	.970	.857	.729	.614	.512	.422	.343	.275	.216	.166	.125	.091	.064
	1	.029	.135	.243	.325	.384	.422	.441	.444	.432	.408	.375	.334	.288
	2	.000	.007	.027	.057	.096	.141	.189	.239	.288	.334	.375	.408	.432
	3	.000	.000	.001	.003	.008	.016	.027	.043	.064	.091	.125	.166	.216
8	0	.923	.663	.430	.272	.168	.100	.058	.032	.017	.008	.004	.002	.001
	1	.075	.279	.383	.385	.336	.267	.198	.137	.090	.055	.031	.016	.008
	2	.003	.051	.149	.238	.294	.311	.296	.259	.209	.157	.109	.070	.041
	3	.000	.005	.033	.084	.147	.208	.254	.279	.279	.257	.219	.172	.124
	4	.000	.000	.005	.018	.046	.087	.136	.188	.232	.263	.273	.263	.232
	5	.000	.000	.000	.003	.009	.023	.047	.081	.124	.172	.219	.257	.279
	6	.000	.000	.000	.000	.001	.004	.010	.022	.041	.070	.109	.157	.209
	7	.000	.000	.000	.000	.000	.000	.001	.003	.008	.016	.031	.055	.090
	8	.000	.000	.000	.000	.000	.000	.000	.000	.001	.002	.004	.008	.017

According to the table, the probability is 0.005.

Solution: Finding Binomial Probabilities Using a Table (2 of 2)

- You can check the result using technology.

MINITAB

Probability Density Function

Binomial with $n = 8$ and $p = 0.05$

x	$P(X = x)$
3	0.0054165

So, the probability that exactly four of the eight workers carpool to work is 0.005. Because 0.005 is less than 0.05, this can be considered an unusual event.

Example: Graphing a Binomial Distribution

Sixty-four percent of people who have survived cancer are ages 65 years or older. You randomly select six people who have survived cancer and ask them whether they are 65 years of age or older. Construct a probability distribution for the random variable x . Then graph the distribution. (Source: American Cancer Society)

Solution:

- $n = 6, p = 0.64, q = 0.36$
- Find the probability for each value of x

Solution: Graphing a Binomial Distribution (1 of 2)

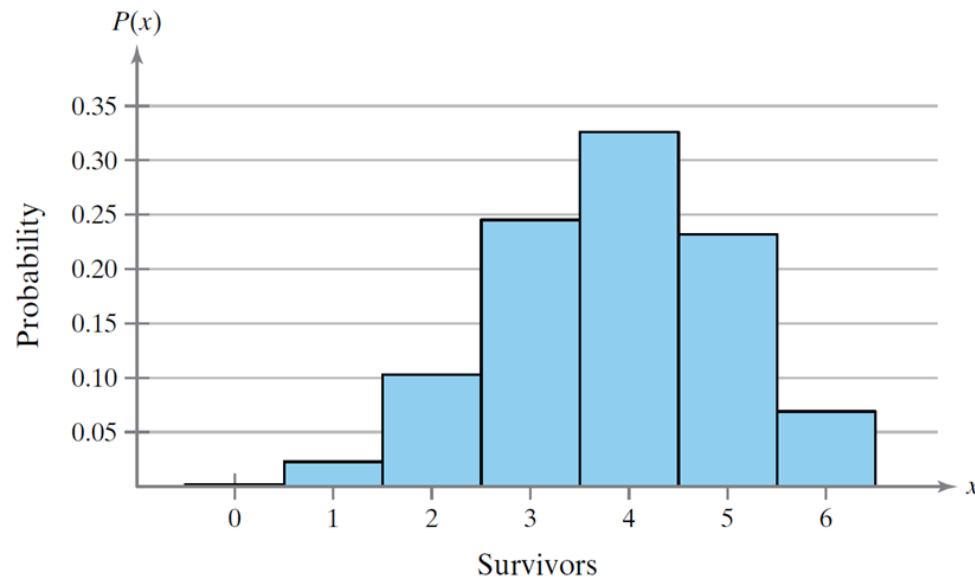
x	0	1	2	3	4	5	6
$P(x)$	0.002	0.023	0.103	0.245	0.326	0.232	0.069

Notice in the table that all the probabilities are between 0 and 1 and that the sum of the probabilities is 1.

Solution: Graphing a Binomial Distribution (2 of 2)

Histogram:

**People Who Have Survived
Cancer, 65 Years of Age or Older**



From the histogram, you can see that it would be unusual for none or only one of the survivors to be age 65 years or older because both probabilities are less than 0.05.

Mean, Variance, and Standard Deviation

- **Mean:** $\mu = np$
- **Variance:** $\sigma^2 = npq$
- **Standard Deviation:** $\sigma = \sqrt{npq}$

Example: Mean, Variance, and Standard Deviation

In Pittsburgh, Pennsylvania, about 56% of the days in a year are cloudy. Find the mean, variance, and standard deviation for the number of cloudy days during the month of June. Interpret the results and determine any unusual values. (Source: National Oceanic and Atmospheric Administration)

Solution: $n = 30$, $p = 0.56$, $q = 0.44$

Mean: $\mu = np = 30 \cdot 0.56 = 16.8$

Variance: $\sigma^2 = npq = 30 \cdot 0.56 \cdot 0.44 \approx 7.4$

Standard Deviation: $\sigma = \sqrt{npq} = \sqrt{30 \cdot 0.56 \cdot 0.44} \approx 2.7$



Solution: Mean, Variance, and Standard Deviation

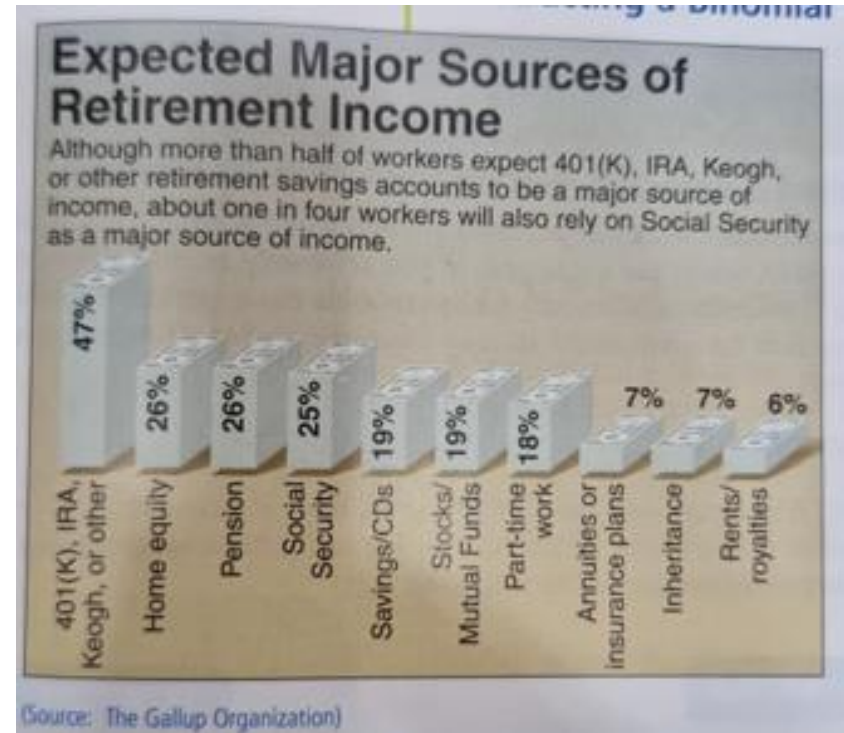
$$\mu = 16.8 \quad \sigma^2 \approx 7.4 \quad \sigma \approx 2.7$$



- On average, there are 16.8 cloudy days during the month of June.
- The standard deviation is about 2.7 days.
- Values that are more than two standard deviations from the mean are considered unusual.
 - $16.8 - 2(2.7) = 11.4$; A June with 11 cloudy days or less would be unusual.
 - $16.8 + 2(2.7) = 22.2$; A June with 23 cloudy days or more would also be unusual.

Exercise 1: Constructing a Binomial Distribution

In a survey, workers in the United States were asked to name their expected sources of retirement income. The results are shown in the survey are randomly selected and asked whether they expect to **rely on Social Security** for retirement income. Create a binomial probability distribution for the number of workers who respond yes.



Solution 1: Constructing a Binomial Distribution

Solution 1: Constructing a Binomial Distribution

Solution 1: Constructing a Binomial Distribution

<i>x</i>	<i>P(x)</i>
	$\sum P(x) =$

Exercise 2: Finding Binomial Probabilities Using Formulas

A survey indicates that 41% of women in the United States consider reading their favorite leisure-time activity. Find the probability that (1) exactly two of them respond yes, (2) at least two of them respond yes, and (3) fewer than two of them respond yes. (Source: Louis Harris & Associate)

Exercise 2: Finding Binomial Probabilities Using Formulas

Exercise 2: Finding Binomial Probabilities Using Formulas

Exercise 3: Graphing a Binomial Distribution

Fifty-nine percent of households in the United States subscribe to cable TV. You randomly select six households and ask each if they subscribe to cable TV. Construct a probability distribution for the random variable x . Then graph the distribution. (Source: Kagan Research, LLC)

Solution 3: Graphing a Binomial Distribution

Solution 3: Graphing a Binomial Distribution

Solution 3: Graphing a Binomial Distribution

Histogram: